1.5.2 Transmission line conductors

Electric power is transmitted from power plant to the load (consumers) through conductors. Copper, aluminum, steel, Aluminum Conductor with Steel Reinforced (ACSR), and cadmium copper are invariably used for transmission line conductors.

1.5.3 Properties of conductors

i. High electrical conductivity
ii. High tensile strength
iii. Low cost
iv. Less weight

The properties of copper and aluminium conductors are discussed below.

a) Copper

Copper is used to transmit large quantity of electricity from one place to another. Hard-drawn copper is often used for power transmission, because it is twice as strong as soft drawn copper. Hard drawn has a high conductivity. Lesser cross-sectional area of conductor is reduced, because the current density of copper is high. It has low specific resistance. Its life is very long.

b) Aluminium

Aluminium conductor is next to copper in its conductivity. It is cheaper than copper and lighter in weight. But conductivity of aluminium is 60% as that of copper. Its diameter is about 1.27 times thicker than that of copper. The melting point of aluminium conductor is less. Hence it creates short circuit.

c) ACSR (Aluminium Conductor with Steel Reinforced)

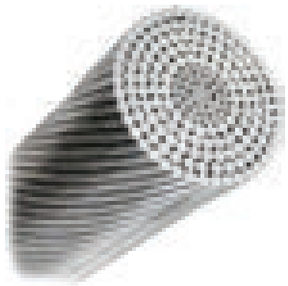


Fig. 1.3 ACSR conductor

An aluminium conductor having a central core of galvanized steel wires is used for high voltage transmission purposes as shown in figure 1.3. This conductor is mostly used for power transmission because, it has less sag and high tensile strength of steel and conductivity of aluminium.

1.6 Types of poles

Line supports or the poles which hold the conductors to a height they are above the ground level. In general, four types of poles are used, depending on the size and shape of the conductor used.

i. Wooden poles
ii. Concrete poles
iii. Tubular steel poles
iv. Latticed steel tower

i. Wooden poles



Fig. 1.4 Wooden poles

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